

Wisdom Science: Measuring, Explaining, and Engineering AI Systems That Improve After Failure

Mian Zhang

Abstract

AI evaluation is dominated by first-attempt performance: accuracy, pass rate, success rate, and benchmark rank. Deployment, however, depends on a different property: whether a system becomes better, more stable, and more transferable after failure, feedback, perturbation, and repeated experience. We call this research program *Wisdom Science*. It separates first-round intelligence from longitudinal wisdom, introduces Wisdom Quotient (WQ) and Embodied Wisdom Quotient (EWQ), studies architecture-dependent wisdom scaling, and requires evidence gates for empirical claims. This paper consolidates the current Wisdom Science package: SOVEREIGN, Cognitive Immunity, WisdomBench, the Intelligence-Wisdom Gap, the Second Scaling Law, WisdomBench-Embodied, Representation Genesis, Embodied Failure Immunity, the Wisdom Science Formal Core, and the Physics/Engineering Core. We provide reproducible artifacts including a claim registry, a terminology registry, a Zenodo bibliography, a macro-mining pass over 130 files, a representation-compression study, formal sanity checks, physics/engineering checks, and an evidence index.

1 The Shift

The dominant question in AI evaluation is: how well does the system perform on the first try? This question is necessary but incomplete. A system deployed in the real world encounters repeated tasks, changing environments, failures, adversarial pressure, and opportunities for transfer. The relevant question becomes:

After experience, does the system become better, more stable, and more transferable?

Wisdom Science is the study of this second question. It does not replace intelligence evaluation. It adds a longitudinal axis.

2 Core Definitions

Definition 1 (First-round Intelligence). *First-round intelligence I is the performance of an agent before task-specific feedback from the current evaluation sequence. For embodied agents we write I_E .*

Definition 2 (Wisdom). *Wisdom is observable improvement conditioned on repeated experience, feedback, failure, and transfer opportunities. It is not a claim about consciousness, virtue, or alignment.*

Definition 3 (Wisdom Quotient). *For N tasks and R rounds, define*

$$\text{WQ} = \frac{1}{N} \sum_{i=1}^N w_i, \quad w_i = \begin{cases} \frac{s_i^{(R)} - s_i^{(1)}}{s_{\max} - s_i^{(1)}} & s_i^{(1)} < s_{\max} \\ 0 & s_i^{(1)} = s_{\max}. \end{cases}$$

WQ measures normalized cross-round learning gain.

Definition 4 (Embodied Wisdom Quotient). *For K embodied tasks, define*

$$\text{EWQ}(M) = \frac{1}{K} \sum_{t=1}^K [s_R(t) - s_1(t)].$$

EWQ measures auditable improvement across repeated physical or simulated embodied trials.

3 Formal Guardrails

The Formal Core separates definitions, theorems, hypotheses, and empirical claims. It proves four guardrails that the rest of the package must obey. First, under non-regression and bounded scores, WQ lies in $[0, 1]$. Second, first-round intelligence and wisdom can rank systems differently, so a first-attempt leaderboard cannot determine a wisdom leaderboard. Third, positive WQ or EWQ does not by itself identify a learning mechanism: the same score trace can be produced by genuine feedback use or by hidden round scheduling. Fourth, aggregate wisdom rankings can reverse if task mixtures are not fixed, so task weights and strata must be reported.

These results are intentionally restrictive. They make the framework harder to overclaim and easier to defend.

4 Physics and Engineering Grounding

The Physics/Engineering Core turns the main variables into operational quantities. Plasticity is treated as normalized response to useful experience. Homeostasis is inverse perturbation gain. Failure immunity is repeat-failure attenuation with related and unrelated controls. Wisdom efficiency is gain per compute, wall-time, API, storage, or robot-wear cost.

The engineering rule is simple: a result is not strong evidence unless it passes metric, stability, provenance, cost, and duplicate-cell gates. This prevents physics language from becoming metaphor and prevents engineering language from becoming presentation polish.

5 The Second Scaling Law

The central hypothesis is that wisdom scales with architectural qualities and experience:

$$\mathbb{E}_{\mathcal{T}}[W(M)] = C_{\mathcal{T}} \Phi(\mathcal{A})^{\alpha} \Psi(\mathcal{A})^{\beta} H(\mathcal{A})^{\gamma} |\mathcal{E}|^{\delta}.$$

Here Φ is plasticity, Ψ is transferable failure immunity, H is homeostasis, and $|\mathcal{E}|$ is the amount of experience. The embodied analogue is

$$W_E \propto \Phi_E^{\alpha} \Psi_E^{\beta} H_E^{\gamma}.$$

Remark 1. *This law is a testable framework, not a finished universal exponent theorem. Current evidence supports the separation between first-round competence and longitudinal improvement, while larger architecture sweeps are required to estimate universal exponents.*

6 The Research Stack

Wisdom Science is organized as a stack rather than a single paper:

Layer	Artifact	Role
System	SOVEREIGN	Cognitive operating system for self-evolving agents
Formal	Formal Core	Definitions, theorems, identifiability, and evidence gates
Physics	Physics/Engineering Core	State dynamics, stability, cost, and acceptance tests
Mechanism	Cognitive Immunity	Failure-to-antibody learning
Benchmark	WisdomBench	Longitudinal text/agent evaluation
Evidence	Six-Model API Panel	Low-cost cross-model longitudinal pilot
Theory	Intelligence-Wisdom Gap	Separates first-round ability from wisdom
Law	Second Scaling Law	Architecture-dependent wisdom scaling
Embodiment	WB-E	Longitudinal robot/VLA evaluation
Evidence	WB-E in Practice	Provenance discipline for embodied claims
Representation	Representation Genesis	Better language for future thought
Grounding	Perspectival Grounding	Views are projections, not the world
Embodied	Embodied Failure Immunity	Failure trajectories as transferable antigens
Organism	Whole-Organism Intelligence	Embodied wisdom beyond a single brain policy

Table 1: Wisdom Science as a layered research stack.

7 Evidence Discipline

Wisdom claims are easy to overstate. For that reason, the package uses claim registries and evidence gates. A claim should name its evidence source, strength, and limitation. For embodied evaluation, every result should point to raw logs, trajectory records, simulator metadata, checkpoint provenance, generated tables, and scripts. For text-agent evaluation, the six-model API panel records raw responses, prompt/response hashes, judge identity, fixed task IDs, seeds, rounds, and summary tables without storing credentials. Its current pilot strengthens cross-model coverage, but it cannot be used as robot rollout or public VLA checkpoint evidence.

8 Representation as the Next Layer

Representation Genesis asks whether a wise system can invent better language for future tasks. The initial macro-mining pass scanned 130 paper and code files. The strongest repeated terms are shown in Table 3.

We also ran a deterministic representation-compression study over 12 core source files and 13 canonical macros. After paying glossary cost, the macro-coded corpus saved 3079 bytes with a compression ratio of 1.0126. This is modest by design. It is a sanity check that the research language is measurably reusable, not a claim that compression alone equals wisdom.

9 Embodied Failure Immunity

Embodied Failure Immunity applies Cognitive Immunity to robot failures. The key object is an embodied antigen: a compact, evidence-backed description of a failure event, causal cue, recovery opportunity, transfer group, and provenance. The first atlas schema includes failure classes such as grasp/contact failure, spatial placement failure, visual grounding failure, instruction grounding failure, perturbation fragility, recovery-loop failure, and simulator/provenance failure.

Table 2: Machine-generated claim registry excerpt.

ID	Object	Evidence
C1	I/WQ	P2, P3
C2	WQ	P1, P2
C3	IWG	P3, P4
C4	SSL	P4
C5	CI	P1
C6	I_E/EWQ	P5, P6, P7
C7	Phi_E/Psi_E/H_E	P5, P6
C8	Gate	P7

Table 3: Top seed terms in the Wisdom Science corpus.

Term	Count	Doc. freq.
immunity	502	33
WisdomBench	178	24
Cognitive Immunity	144	27
Ouroboros	106	31
homeostasis	75	19
plasticity	64	14
Second Scaling Law	52	13
Wisdom Quotient	49	22
Cognitive Entropy	40	9
Intelligence-Wisdom Gap	35	12
Evidence Gate	32	13
WisdomBench-Embodied	26	10

P9 now includes a small matched recovery-adaptor pilot: eight public-factory baseline failures, three rescues, and no regressions. The result is deliberately bounded; it supports an antigen-derived non-training recovery pilot, not a public SOTA leaderboard or a trained-policy claim.

10 Views and Organisms

Two additional principles complete the current stack. First, *the view is not the world*: a single observation may correspond to many latent states, meanings, or strategic intentions. Perspectival Grounding therefore requires agents to preserve competing hypotheses and select additional views, actions, context, or belief-model evidence before collapsing ambiguity. Second, *the brain is not the organism*: embodied wisdom depends on policy, senses, routing, metabolic budget, joints, immune memory, calibration, social norms, and counter-reflexive game cognition. Whole-Organism Intelligence turns those organ-like subsystems into engineering gates rather than biological metaphors.

11 Leaderboard Schema

The proposed leaderboard separates first-round intelligence from longitudinal wisdom:

- Language agents: report I , WQ , Φ , Ψ , H , generalization, and repeat failure.
- Embodied agents: report I_E , EWQ , Φ_E , Ψ_E , H_E , trajectories, and provenance.
- Representation systems: report compression gain, unpacking cost, transfer, reuse, novelty, and residual absorption.

Table 4: Initial embodied failure atlas classes.

Failure class	Typical antigen cue
Grasp Contact Failure	missed contact, slip, unstable grasp, collision
Placement Spatial Failure	wrong pose, wrong target, failed insertion
Occlusion Visual Grounding Failure	object hidden, visual aliasing, distractor
Perspectival Aliasing Failure	one projection supports multiple latent states
Active Disambiguation Failure	agent fails to gather next view, touch, or context
Instruction Grounding Failure	wrong object, wrong relation, semantic ambiguity
Perturbation Fragility	lighting, friction, position, or dynamics shift
Sensor Pipeline Failure	sensor stream missing, stale, misrouted, or uncalibrated
Actuator Joint Instability	unstable joint, slip, saturation, or controller lag
Metabolic Budget Failure	battery, latency, GPU, API, or memory budget exhausted
Psychological Calibration Failure	overconfidence, panic loop, refusal collapse
Social Norm Grounding Failure	physical success violates role, norm, or trust boundary
Counter Reflexive Game Failure	missed bluff, feint, recursive belief, or opponent adaptation
Recovery Loop Failure	repeated retry without new recovery behavior
Simulator Or Provenance Failure	missing trajectory, checkpoint, metadata, or seed
Unknown	insufficient evidence for classification

This schema prevents a common leaderboard error: treating a high first attempt as evidence of learning.

12 Limitations

Wisdom Science is not a claim that benchmark improvement equals moral wisdom. WQ and EWQ are operational metrics. Positive WQ or EWQ may arise from stochasticity, hidden state, task ordering, or simulator artifacts. Therefore, mechanism-level provenance is required. The Second Scaling Law remains a framework with preliminary evidence, not a universal exponent theorem. Representation compression is a useful signal but not truth.

13 Conclusion

Wisdom Science adds a second axis to AI evaluation. The first axis asks how competent a system is before experience. The second asks whether it becomes better after experience. The resulting research program connects benchmarks, architecture, failure learning, embodied evaluation, representation search, and evidence discipline. The goal is not merely to build larger models, but to build systems that can fail, learn, transfer, stabilize, and improve.

References

- [1] Mian Zhang. SOVEREIGN: A Cognitive Operating System for Self-Evolving AI Agents. Zenodo, 2026.
- [2] Mian Zhang. WisdomBench: A Longitudinal Benchmark for Measuring Wisdom Acquisition in AI Agents. Zenodo, 2026.

- [3] Mian Zhang. The Intelligence-Wisdom Gap: Why Smarter AI Agents Are Not Wiser Ones. Zenodo, 2026.
- [4] Mian Zhang. The Second Scaling Law: Intelligence Scales with Parameters, Wisdom Scales with Architecture. Zenodo, 2026.
- [5] Mian Zhang. WisdomBench-Embodied. Zenodo, 2026.
- [6] Michael H. Freedman et al. Compression is all you need: Modeling Mathematics. arXiv:2603.20396, 2026.